

Graphs of Rational Functions (2.8)

Name _____

Circuit

Begin with the problem in the top left. Answer the question. To advance in the circuit, find your answer and write “2” in the blank. Solve that problem, find your answer and write “3” in the blank and so on until you have solved every problem and returned to the beginning.

ANSWER: no # _____ Find the slant asymptote of $f(x) = \frac{x^2 + x}{x - 1}$.	ANSWER: 1 # _____ State the x-intercept of $f(x) = \frac{x + 3}{x - 1}$.
ANSWER: $x = 2$ # _____ State the horizontal asymptote of $f(x) = \frac{x^2}{x^2 - x - 2}$.	ANSWER: $x = -1, x = 2$ # _____ Can a rational graph cross its horizontal asymptote?

ANSWER: $y = x + 2$ # _____ State the vertical asymptote of $f(x) = \frac{3}{x-2}$.	ANSWER: $y = x$ # _____ State the vertical asymptotes of $f(x) = \frac{x^2}{x^2 - x - 2}$.
ANSWER: $(0, -\frac{3}{2})$ # _____ A rational function has a slant asymptote when the numerator degree is how much more than the denominator's?	ANSWER: yes # _____ Can a rational graph cross a vertical asymptote?
ANSWER: $y = 1$ # _____ State the y-intercept of $f(x) = \frac{3}{x-2}$.	ANSWER: $(-3, 0)$ # _____ Find the slant asymptote of $f(x) = \frac{x^3}{x^2 - 4}$.